

Version 1 - solution

1. [7 marks] Use the rules of inference to show that the following argument is valid. For each step, state the rule used and the step(s) to which it applies (if applicable).

$$p \vee q \vee r$$

$$u \wedge v \wedge \neg w$$

$$q \rightarrow (\neg u \wedge r)$$

$$(r \rightarrow s) \wedge (t \rightarrow q)$$

$$\therefore \neg t$$

direct
proof

proof
by
contrad.

STEPS		REASON
1	$u \wedge v \wedge \neg w$	given
2	u	C.S. 1
3	$u \vee \neg r$	D.A. 2
4	$\neg(\neg u \wedge r)$	DeMorgan's 3
5	$q \rightarrow (\neg u \wedge r)$	given
6	$\neg q$	M.T. 4,5
7	$(r \rightarrow s) \wedge (t \rightarrow q)$	given
8	$t \rightarrow q$	C.S. 7
9	$\neg t$	M.T. 6,8
10	t	negate concl.
11	$(r \rightarrow s) \wedge (t \rightarrow q)$	given
12	$t \rightarrow q$	C.S. 11
13	q	M.P. 10,12
14	$q \rightarrow (\neg u \wedge r)$	given
15	$\neg u \wedge r$	M.P. 13,14
16	$\neg u$	C.S. 15

$$17 \quad u \wedge v \wedge \neg w$$

$$18 \quad u$$

$$19 \quad \neg t$$

Lines 16, 18 are in Contrad.

2. [2 marks] Give a set of specific values for p , q , and r that shows that the following argument is **invalid**. Justify your answer.

$$p \wedge q \quad 1 \wedge 1 = 1$$

$$p \vee (q \rightarrow r) \quad 1 \vee (1 \rightarrow 1) = 1$$

$$(\neg q \vee r) \rightarrow p \quad (0 \vee 1) \rightarrow 1 = 1$$

$$\therefore \neg(r \vee q) = \neg r \wedge \neg q \quad 0 \wedge 0 = 0$$

$$p = 1; q = 1; r = 1$$

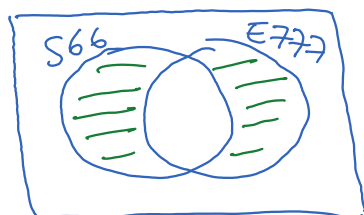
Makes all premises true
and the conclusion false

3. [3 marks] What is the probability that a randomly selected **octal** string of length 10 starts with 66 or ends with 777, but not both?

$$\begin{array}{c} \underline{6} \quad \underline{6} \quad \underline{\quad} \quad \underline{\quad} \quad \underline{\quad} \quad \underline{\quad} \quad \underline{\quad} \quad \underline{\quad} \quad \underline{\quad} \quad \underline{\quad} \\ (8) \quad (8) \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad (8) \end{array} \quad 8^8$$

$$\begin{array}{c} \underline{\quad} \quad \underline{\quad} \quad \underline{\quad} \quad \underline{\quad} \quad \underline{\quad} \quad \underline{7} \quad \underline{7} \quad \underline{7} \\ (8) \quad (8) \quad \cdot \quad \cdot \quad \cdot \quad \quad \quad (8) \end{array} \quad 8^7$$

$$\begin{array}{c} \underline{6} \quad \underline{6} \quad \underline{\quad} \quad \underline{\quad} \quad \underline{\quad} \quad \underline{7} \quad \underline{7} \quad \underline{7} \\ (8) \quad (8) \quad (8) \quad \quad \quad (8) \quad (8) \end{array} \quad 8^5$$



$$Pr = \frac{(8^8 - 8^5) + (8^7 - 8^5)}{8^{10}}$$

4. [3 marks] A student has 40 identical Halloween treats.

a) In how many ways can she distribute them to 5 different friends?

$$x_1 + x_2 + x_3 + x_4 + x_5 = 40 ; x_1 \dots x_5 \geq 0$$

$$\begin{matrix} n=5 \\ r=40 \end{matrix} \quad \binom{n+r-1}{r} = \binom{5+40-1}{40} = \binom{44}{40}$$

b) How many distributions in part a) have one friend receiving more than half of the treats?

$$x_1 + x_2 + x_3 + x_4 + x_5 = 40$$

Making sure that x_1 gets at least 21 treats, giving

$$x_1 \text{ 21 treats first ; } r = 40 - 21 = 19$$

$$\begin{matrix} n=5 \\ r=19 \end{matrix} \quad \leftarrow \begin{matrix} \text{distributing} \\ \text{remaining treats} \end{matrix}$$

$$\binom{n+r-1}{r} = \binom{5+19-1}{19} = \binom{23}{19}$$

$$\therefore \binom{23}{19} \cdot 5$$

that friend can
be $x_1, x_2, \dots, x_5$

5. [9 marks] The System Administrator has set the following rules for the password:

- ✓ The password is a string of length 10
- ✓ Only the following 36 characters can be used:

26 letters: A, B, C, D, ..., X, Y, Z
 10 digits: 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9

- ✓ Repetition is allowed unless otherwise stated

- a) [3 marks] How many different passwords, with three digits and seven letters, have no adjacent digits?

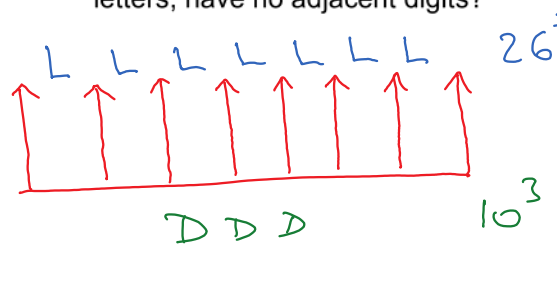


Diagram showing 7 letter slots (L) and 3 digit slots (D) with arrows indicating no adjacent digits. The calculation is:

$$\binom{8}{3} \cdot (26)^7 \cdot (10)^3$$

where $\binom{8}{3}$ is the number of ways to place the 3 digits among the 8 possible gaps (before, between, and after the 7 letters).

- b) [3 marks] How many different passwords have at least four letters and at least four digits?

4L and 6D OR 5L and 5D OR 6L and 4D
 L L L L D D D D D D OR L L L L L D D D D D OR L L L L L L D D D D

$$\frac{10!}{4!6!} \cdot (26)^4 (10)^6 + \frac{10!}{5!5!} (26)^5 (10)^5 + \frac{10!}{6!4!} \cdot (26)^6 (10)^4$$

- c) [3 marks] How many different passwords, with letters only, have some letter present four times, a different letter present three times, and three other distinct letters? 문자만 있는 암호에 어떤 문자는 4번, 다른 문자는 3번, 다른 3개의 다른 문자가 있는 암호는 몇 개입니까?

Diagram showing the selection of letters: $L_1, L_1, L_1, L_1, L_2, L_2, L_2, L_3, L_4, L_5$

$$\frac{10!}{4!3!} \cdot \binom{26}{1} \binom{25}{1} \binom{24}{3}$$

where $\binom{26}{1}$ is the number of ways to select L_1 , $\binom{25}{1}$ is the number of ways to select L_2 , and $\binom{24}{3}$ is the number of ways to select L_3, L_4, L_5 .

6. [4 marks] Circle one answer only; you MUST justify the answer. No marks will be given for answers without justification. No marks will be given when justification does not imply the correctness of the answer.

Universe $U = \{-6, -5, -3, -2, -1, 0, 1, 2, 3, 5\}$

$X = \{-3, -2, 1, 2, 3\}$ $Y = \{-2, -1, 1, 2\}$ $Z = \{-6, -5, -2, 0, 1, 2, 6\}$

a) $\exists x \in X, \forall y \in Y$ then $x \cdot y \notin Z$

TRUE / FALSE

x	y	$x \cdot y \notin Z$
-3	-2	6 $\notin Z$ x
-2	-1	2 $\in Z$ x
1	1	1 $\in Z$ x
2	1	2 $\in Z$ x
3	2	6 $\in Z$ x

$Y \Delta Z = \{-2, -1, 1, 2\} \Delta \{-6, -5, -2, 0, 1, 2, 6\} = \{-6, -5, -1, 0, 6\}$

$X \Delta Y = \{-3, -2, 1, 2, 3\} \Delta \{-2, -1, 1, 2\} = \{-3, -1, 3\}$

b) $\exists x \in (Y \Delta Z), \forall y \in (X \Delta Y)$ then $x \cdot y \in Z$
 $\{-6, -5, -1, 0, 6\}$ $\{-3, -1, 3\}$

TRUE / FALSE

x	y	$x \cdot y \in Z$
0	-3	0 $\in Z$ ✓
0	-1	0 $\in Z$ ✓
0	3	0 $\in Z$ ✓

7. [4 marks] Consider the following programming segment. Your answer must rely on combinations structure. Answers that use sigma notation will not be accepted.

Determine the value of the variables `counter1`, and `counter2` after the segment is executed.

Do not simplify the expressions.

(Note: the for loops include both the start and end values)

```

counter1 = 50
counter2 = 100
for i = a+2 to 2*n do {
    counter1 = counter1 + 11
    for j = i+1 to 5*n do {
        counter2 = counter2 + 22
    }
}
//assume a and n are positive integers with n>a

```

Answer:

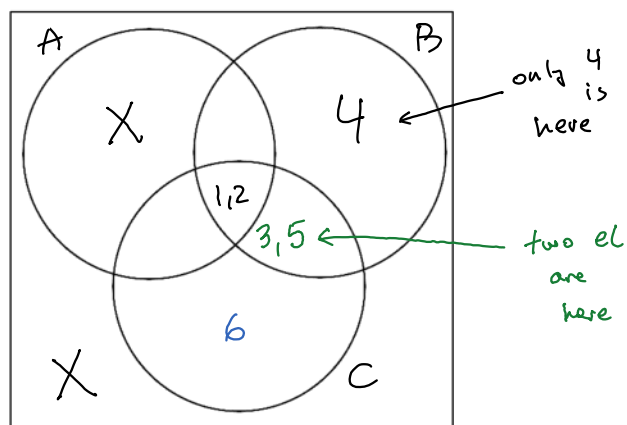
$$\text{counter1} = 50 + (2n - (a+2) + 1)(11) = 50 + (2n - a - 1)(11)$$

$$\text{counter2} = 100 + \left[\binom{5n - (a+2) + 1}{2} - \binom{5n - 2n}{2} \right] (22)$$

$$= 100 + \left[\binom{5n - a - 1}{2} - \binom{3n}{2} \right] (22)$$

8. [5 marks] Provide one example of nonempty sets A, B and C that satisfies all of the following:

- $(A - B) \subseteq C$
- $B - (A \cup C) = \{4\}$
- $\{1, 2\} \subseteq \overline{(A \cap B) \Delta C}$
- $\overline{(C - A)} \subseteq (A \cup B)$
- $|\overline{A \cup (\overline{B} \cap C)}| = 3$



Universe $U = \{1, 2, 3, 4, 5, 6\}$.

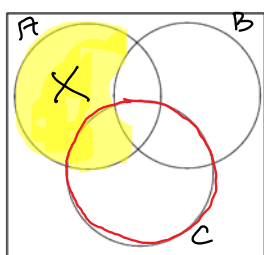
Write your answer: *one solution is*

$$A = \{1, 2\}$$

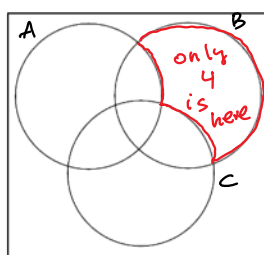
$$B = \{1, 2, 3, 4, 5\}$$

$$C = \{1, 2, 3, 5, 6\}$$

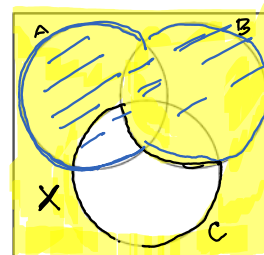
Must write your answer here!
Leaving at least one set empty
implies zero marks.



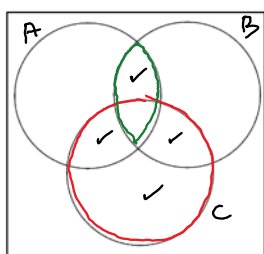
$$(A - B) \subseteq C$$



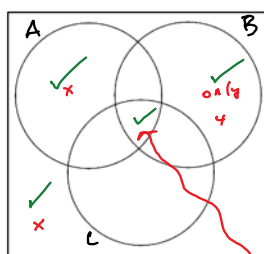
$$B - (A \cup C) = \{4\}$$



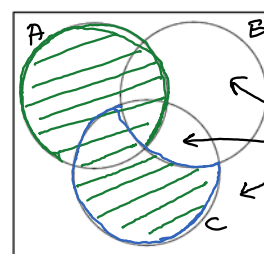
$$\overline{(C - A)} \subseteq (A \cup B)$$



$$(A \cap B) \Delta C$$



$$\{1, 2\} \subseteq (A \cup B) \cap C$$



$$A \cup (B \cap C)$$

$\overline{A \cup (B \cap C)}$
3 el. are here

Complement
1, 2 are here.
see previous figures